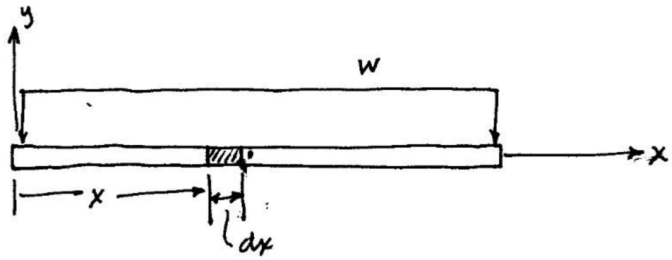
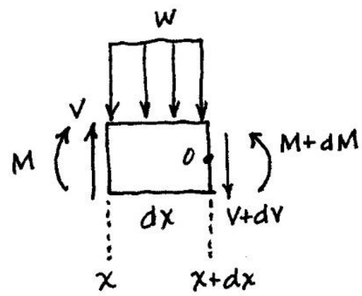
CE 2200 Statics: Internal Forces, Deriving Shear and Bending Moment Relationships

DERIVATION OF SHEAR & BENDING MOMENT RELATIONSHIPS



LOOKING AT ELEMENT



$$+\uparrow \Sigma F_y = 0; \quad \cancel{V} - w\,dx - (\cancel{V} + dV) = 0$$

$$dV = -w\,dx$$

$$\boxed{\frac{dV}{dx} = -w}$$

OR

$$\int_{V_1}^{V_2} dV = -\int_{X_1}^{X_2} w\,dx \quad \Big\} \text{ Sep. Vars}$$

$$\boxed{V_2 - V_1 = \Delta V = -\int_{X_1}^{X_2} w(dx)}$$

$$\circlearrowleft + \Sigma M_o = 0; \quad -\cancel{M} - V\,dx + \underbrace{(w\,dx)\frac{dx}{2}}_{\text{Negligible}} + (\cancel{M} + dM) = 0$$

$$dM = V\,dx$$

$$\boxed{\frac{dM}{dx} = V}$$

$$\int_{M_1}^{M_2} dM = \int_{X_1}^{X_2} V\,dx \quad \Big\} \text{ Sep. Vars}$$

$$\boxed{M_2 - M_1 = \Delta M = \int_{X_1}^{X_2} V\,dx}$$